AI Camera Vision Analytics & SaaS Tenant Management Platform

Complete Technical Blueprint & Operational Walkthrough | System Architecture & Computer Vision AI Models Guide

1. System Architecture & Tech Stack

The platform is structured as a decoupled full-stack Web application powered by Vite + React on the frontend and Django REST Framework backed by PostgreSQL on the server layer.

Component	Technology	Details & Specification
Frontend	Vite + React 18 + TS	TypeScript, Custom Vanilla CSS Token System, Lucide Icons, Canvas Crop Engine
Backend API	Python 3.14 + Django 5	Django REST Framework, CORS Headers (`django-cors-headers`), Port 8001
Database	PostgreSQL 16+	Database: `ipcamera`, User: `postgres`, Port 5432, Enforced strict ORM sync
Computer Vision	ONVIF Profile T / RTSP	4K / 1080p @ 30 FPS streams with real-time Edge AI bounding box overlays

2. Computer Vision AI Models (Face & Object Identification)

The platform integrates specialized deep learning model architectures tuned for real-time video stream analytics, face identification, and object detection:

AI Capability / Module	Model Architecture	Framework / Backbone	Function & Accuracy
Facial Recognition Kiosk	ArcFace	ResNet-100 (512-d embeddings)	High-precision staff clock-in recognition (99.4% Acc)
Face Detection & Alignment	RetinaFace	MobileNetV2 Feature Pyramid	Face localization under store lighting (98.7% Acc)
People & Dwell Tracking	YOLOv8-PersonTrack-v2.1	YOLOv8x + ByteTrack	Foot-traffic counting & dwell heatmaps (94.2% Acc)
Produce & Bin Segmentation	ShelfSeg-ProduceNet-v1.4	Mask R-CNN / DeepLabV3+	Fresh produce fill % estimation (91.8% Acc)
Checkout Queue Depth	QueueDepth-v3.0	YOLOv8 Nano + DeepSORT	Wait-time alerts & queue tracking (89.5% Acc)
Aisle Hazard & Spill Spot	HazardSpot-v1.0	Swin Transformer (Swin-B)	Fallen produce & floor spill alerts (86.0% Acc)
Loss Prevention Action Cues	ActionCue-LPR-v0.9	SlowFast 3D ResNet	Concealment cue flagging for review (83.4% Acc)